DS311: EXPLORATORY DATA ANALYSIS • SECTION DS3A / DS3B • GROUP 4 (ENVIRONMENT & NATURE)

PIT POSTER | A2: 420 × 594 MM | UNSD EMPIRICAL COHORT (n=24)

Analysing the growth of Municipal Wastewater Generation and its Burden on Urban Treatment Infrastructure

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CORE DISCOVERY

Across sovereign nations, every **1,000 m³/day** surge in municipal wastewater generation yields only **296 m³/day** in urban treatment expansion ($\beta_1 = 0.2964$, $R^2 = 82.0\%$) — creating an unabsorbed **70.4% structural infrastructure deficit** that bypasses treatment.

01 Research Problem & Inquiry

SDG 6.3 & 11.6

Accelerated urbanization and demographic concentration drive exponential growth in municipal effluent. When centralized wastewater treatment plants (WWTPs) fail to scale proportionately, untreated discharges directly infiltrate receiving water bodies, triggering severe eutrophication, pathogen contamination, and macro-economic environmental damage.

The Business Invalidation Question: *Does national urban treatment capacity expand at a 1:1 parity with municipal wastewater generation, or does an operational infrastructure gap widen — and what is the quantified capital expenditure (CAPEX) required to close it?*

02 Dataset & Case Extraction

UNSD Registry

Empirical records were procured from the **United Nations Statistics Division (UNSD)** Environmental Indicators (Inland Water Resources).

Data Partition Stage	Records	Methodological Rationale
Raw Global Registry	80 nations	Global reporting database
Pruned Missing Rows	56 (70.0%)	Removed to avoid imputation noise
Complete Bilateral Cases	**24 (30.0%)**	**Rigorous Complete Case Analysis**

Complete bilateral reporting requires simultaneous measurement of generation (Variable X) and treatment (Variable Y) under identical statutory accounting standards.

03 Univariate: Wastewater Generated (X)

n = 24 Nations

Total Municipal Wastewater Generated exhibits extreme right-skewness and high kurtosis, driven by large sovereign populations.

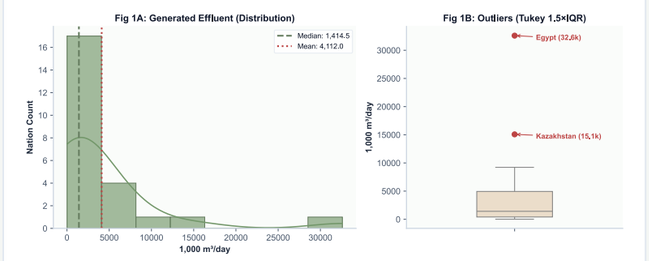


Figure 1: Univariate distribution and Tukey 1.5×IQR outlier fence for Variable X.

Statistic	Value	Interpretation
Median (Q2)	1,414.52 k m³/d	Robust center of distribution
IQR (Q3 - Q1)	4,532.91 k m³/d	Middle 50% dispersion
Skewness	+3.23	Severe positive / right asymmetry
Outlier Fence	11,739.53 k m³/d	Egypt (32.6k), Kazakhstan (15.1k)

04 Macro Empirical Benchmarks

Pearson r & Regression

+0.906
Pearson Correlation (r)
Near-perfect linear coupling ($p < 0.001$)

82.04%
Variance Explained (R^2)
Generation dictates treatment scale

0.2964
Marginal Slope (β_1)
Only 29.6% absorbed per 1,000 m³ surge

70.36%
Infrastructure Deficit
Untreated marginal volume gap

05 Bivariate Scaling vs. 1:1 Parity

Ordinary Least Squares

Plotting Variable Y against Variable X reveals that while the correlation is high, the slope **$\beta_1 = 0.2964$ diverges starkly from the 1:1 Parity line**.

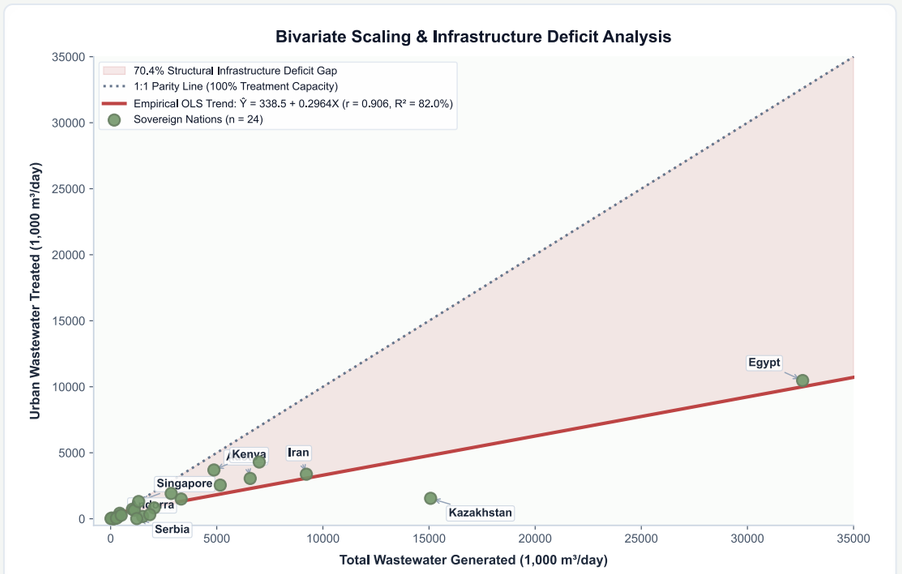


Figure 2 (Hero Visual): Bivariate scatterplot of n=24 reporting nations with empirical OLS trajectory, 1:1 parity line, and shaded structural deficit zone.

Mathematical Model: $\hat{Y} = 338.45 + 0.2964 \times X$ ($p = 1.15 \times 10\text{-}9$)

Sovereign treatment capacity expands at under one-third the rate of generation. Nations above 10,000 k m³/d accumulate massive unabsorbed effluent burdens.

06 Univariate: Wastewater Treated (Y)

Urban Treatment Flow

Urban Wastewater Treated also displays a right-skewed profile, with sovereign treatment clustered at lower volumes.

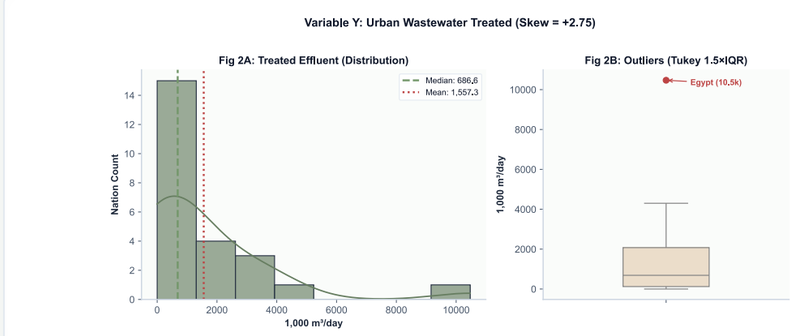


Figure 3: Univariate distribution and Tukey outlier detection for Variable Y.

Statistic	Value	Interpretation
Median (Q2)	686.64 k m³/d	Less than half of median generated volume
IQR (Q3 - Q1)	1,952.88 k m³/d	Narrower capacity distribution
Skewness	+2.75	Constrained positive distribution
Outlier Fence	5,004.99 k m³/d	Egypt (10,470 k m³/d) solitary extreme outlier

07 The 70.4% Allocation Gap

Capacity Shortfall

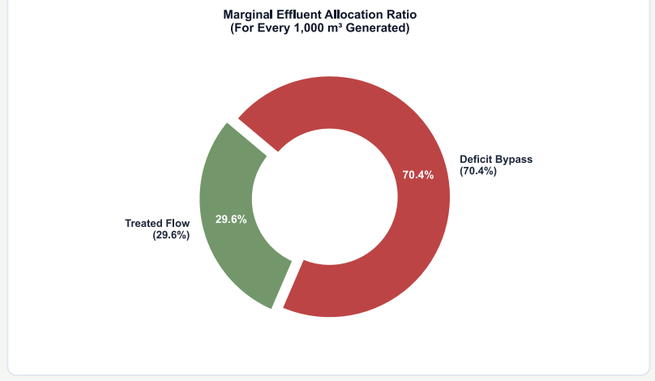


Figure 4: Marginal effluent allocation ratio per 1,000 m³ of new wastewater.

In aggregate across the reporting cohort, **61.3 Million m³/day (22.4 Billion m³/year)** of wastewater remains uncollected or untreated, discharging raw organic loads into receiving river basins and coastal shelves.

08 Economic Valuation & CAPEX

Utility Concessions

$122.6 Billion USD Deficit Market

Benchmark WWTP Construction Cost: ~$2.00 / m³ annual capacity

National Archetype Contrasts:

- **Parity Leaders (100% Treated):** Singapore (99.0%), Andorra (100%), Monaco (100%) demonstrate that high treatment efficiency is technologically and financially viable through circular water reclamation.
- **Critical Deficit Hotspots:** Kazakhstan (10.3% treated, 13.5M m³/d gap), Serbia (1.1% treated), Egypt (32.1% treated, 22.1M m³/d gap) require targeted multinational development bank interventions.

09 Prescriptive Policy Interventions

Engineering Action

1. **Decentralized Modular WWTPs (SBR Systems)**
   Deploy compact Sequencing Batch Reactors near urban satellite clusters rather than waiting for multi-billion dollar centralized pipe networks.
2. **Full-Cost Recovery Utility Tariffs**
   Restructure municipal water pricing to ring-fence sewer surcharges, creating predictable revenue streams for private utility concessions.
3. **Bilateral Data Harmonization**
   Address the 70.0% UNSD reporting attrition by providing technical capacity grants for municipal flow meters in emerging economies.

SCAN FOR REPRODUCIBLE DATA
GitHub: Troge-dev/datasets • EDA Midterms
Cleaned CSV, Jupyter Notebook & Analysis

References & Methodology: United Nations Statistics Division (UNSD), 2021. *Environmental Indicators: Inland Water Resources • Wastewater Generation and Treatment*. Complete Case Analysis (n=24). Developed for DS311 Exploratory Data Analysis Performance Innovative Task (Group 4: Environment & Nature).